CREDIT RISK · PORTFOLIO REVIEW

Credit Risk Portfolio Intelligence

Where the risk sits, which accounts need attention now, and an AI assistant that answers questions from the data, with a person in every decision.

Prepared for the Chief Risk Officer · NITISH MALIK · October 2026

THE PROBLEM

Three gaps in how the book is managed today

01

Concentration hides in averages

The book defaults at 16.1%, yet mortgages, one account in ten, carry 65% of the money lost to defaults.

02

No account-level early warning

Risk is managed by segment. Analysts can’t see which accounts are slipping, or why, until collections starts.

03

Simple questions take days

“Which segment got worse?” goes into an analyst queue and comes back as a spreadsheet nobody can trace.

Source: 5 provided tables, 14,685 loan accounts, 10,000 customers (synthetic)

THE BOOK AT A GLANCE

One in six loans defaults, and the losses are not spread evenly

14,685
loan accounts held by 10,000 customers

$601M
exposure (original loan amount)

16.1%
of accounts defaulted (2,364 loans)

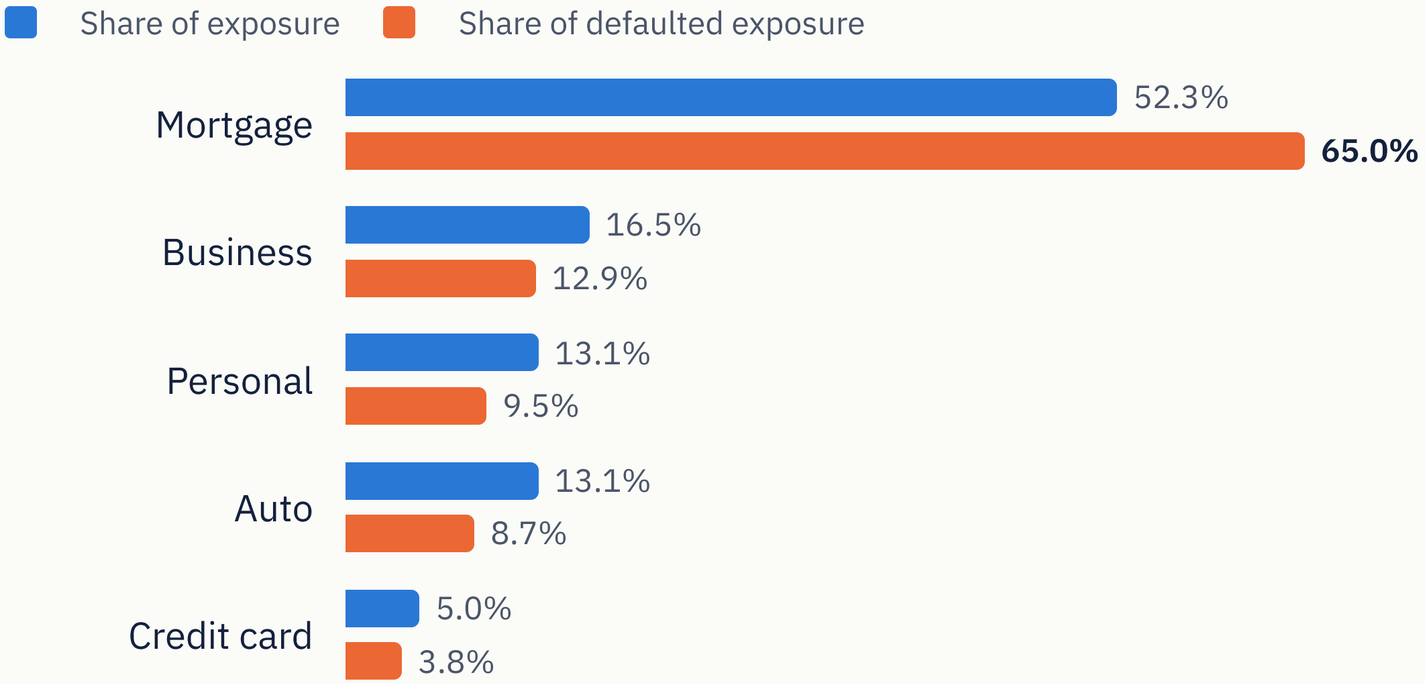
65%
of defaulted exposure sits in mortgages, 10% of accounts

Before any analysis, the data needed work: 45 automated checks found 30 issues, from inconsistent labels to collections events dated before their loans existed. Each one is counted, fixed or flagged, and logged.

Source: accounts_clean.csv; data_quality_notes.md

INSIGHT 1 · CONCENTRATION

Mortgages = 52% of exposure, 53% higher default rate



24.7%

mortgage default rate, 1.5× the book. The median mortgage is 2.8× the borrower's annual income; other products sit at 0.2×.

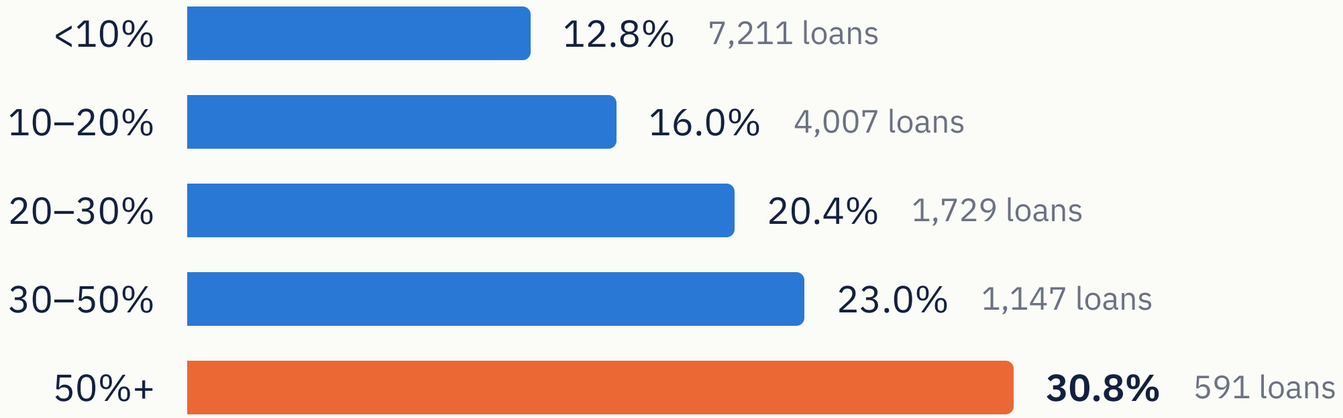
So what: review mortgage underwriting first, and get collateral (LTV) data, which this dataset lacks, to size real losses.

Exposure = original loan amount. Source: notebooks/02_data_analytics.ipynb

INSIGHT 2 · AFFORDABILITY

Instalment at 30%+ of income = 33% of exposure, 59% higher default rate

Default rate by monthly instalment as a share of monthly income (book average 16.1%)



It holds within each product: mortgages go from 17.3% at 10–20% of income to 47.1% above half of income.

79

loans have an instalment larger than the borrower’s whole monthly income. 45.6% of them defaulted.

So what: a hard affordability cap at origination, for example instalment at most 40% of income. Today impossible loans get through.

Source: notebooks/02_data_analytics.ipynb, sections 2.3, 2.4 and 2.6

INSIGHTS 3–5 · WHO DEFAULTS

Low income and unemployment compound; region alone doesn’t

Insight 3 · income

+74%

Income under $50k = 12.7% of exposure, 74% higher default rate (27.9%). Under $30k: 140% higher.

So what: income-tiered limits and pricing; verified income below $50k.

Insight 4 · income × employment

+175%

Unemployed under $30k = 1.6% of exposure, 175% higher default rate (44.3%). Salaried on $100k+: 44% lower.

So what: verified alternative income and manual review.

Insight 5 · region × loan type

+70%

South mortgages = 14.0% of exposure, 70% higher default rate (27.3%). Region alone: 15.0–16.9%.

So what: review that book; no region-level policy.

Higher default rate = vs the 16.1% portfolio average. Source: notebooks/02_data_analytics.ipynb, sections 2.4 and 2.6

THE DEFAULT MODEL

An explainable score for every account

0.687
AUC on customers the model never saw

0.700
AUC out of time, on 2022–23 loans

0.375
Gini on the hold-out test set

0.288
KS on the hold-out test set

The riskiest 30% of accounts hold 54% of all defaults.
Predicted and actual default rates match by band:

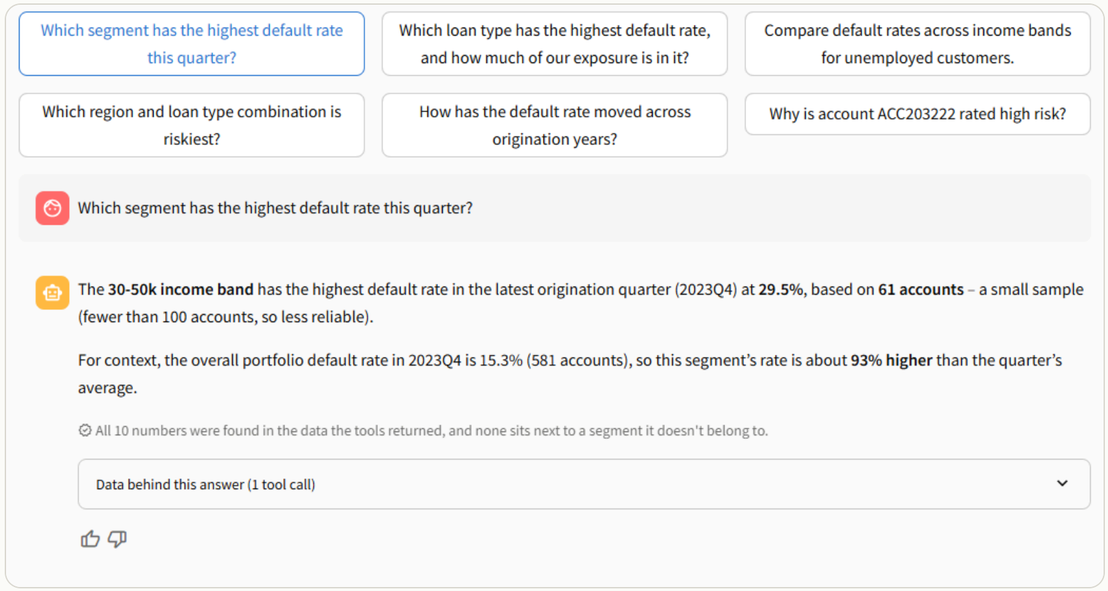
Risk band	Predicted	Actual
Low	7.7%	6.8%
Medium	14.2%	14.4%
High	25.4%	26.1%
Very High	45.5%	41.9%

Logistic regression beat LightGBM (AUC 0.678, and it overfits). We ship the transparent scorecard, whose reasons for each account are exact, and keep LightGBM as the challenger.

Hold-out = 20% of customers, all their loans. Source: notebooks/03_model_building.ipynb, docs/model_report.md

THE AI FEATURE, INSIDE THE DASHBOARD

AI feature: Ask the Portfolio



The brief's own example, "Which segment has the highest default rate this quarter?", answered from 2023 Q4 data: all 10 numbers verified.

Grounded: Nemotron can only query the data through 8 tools; every number is checked against the data and its segment.

What triggers it

An analyst's question, typed or suggested, over the whole book or the dashboard's filters.

False positive

A confident answer the data can't support; queries are shown, and numbers not in the data or next to the wrong segment are flagged.

False negative

"I can't answer that" when the data could; rephrase, or use the same cut in the dashboard.

Human in the loop

The analyst checks the data behind a figure before using it, and rates the answer.

Screen from the working Streamlit dashboard. Details: docs/ai_features.md, notebooks/04_ai_feature.ipynb

RISK AND COMPLIANCE

Built to pass model risk and fair-lending review

Area	Control	Evidence
Model governance (SR 11-7)	Champion and challenger; tested on unseen customers and out of time	Test AUC 0.687, out of time 0.700; LightGBM rejected
Fair lending (ECOA, Reg B)	Gender, marital status and age excluded; adverse impact tested for every group	Gender ratio 0.99; under-26s 0.72, sent for Compliance review
Explainability	Exact reasons for every account; each feature must pass a direction check	Basis for adverse-action reason codes
AI controls	No automated customer action; chat numbers verified; analyst decisions logged	Invented or misplaced numbers flagged in live tests; in-house model hosting for production
Data quality	45 automated checks every run; leakage tests on each feature group	30 issues fixed or flagged; no default visible inside the feature window

Ratio = adverse impact ratio, four-fifths rule (pass at 0.80 or above). Source: docs/model_report.md, docs/assumptions_log.md

RECOMMENDATIONS AND THE ASK

Four policy moves, and a 90-day pilot

POLICY MOVES, FOR CREDIT POLICY TO SIZE

1 · Cap the instalment at 40% of income at origination; above 50%, loans default at 30.8%.

2 · Review mortgage underwriting and collect collateral values: 65% of defaulted exposure.

3 · Verify income under $50k; send unemployed applicants to manual review.

4 · Use Ask the Portfolio in the monthly risk review.

90-DAY PILOT ON THE BANK'S OWN DATA

Month 1, connect: current balances, dated bureau pulls, the bank's default definition; rerun the 45 checks.

Month 2, validate: independent model validation, fair-lending review, security review of in-house AI.

Month 3, pilot: five analysts use the dashboard and chat; measure accuracy, speed and stability.

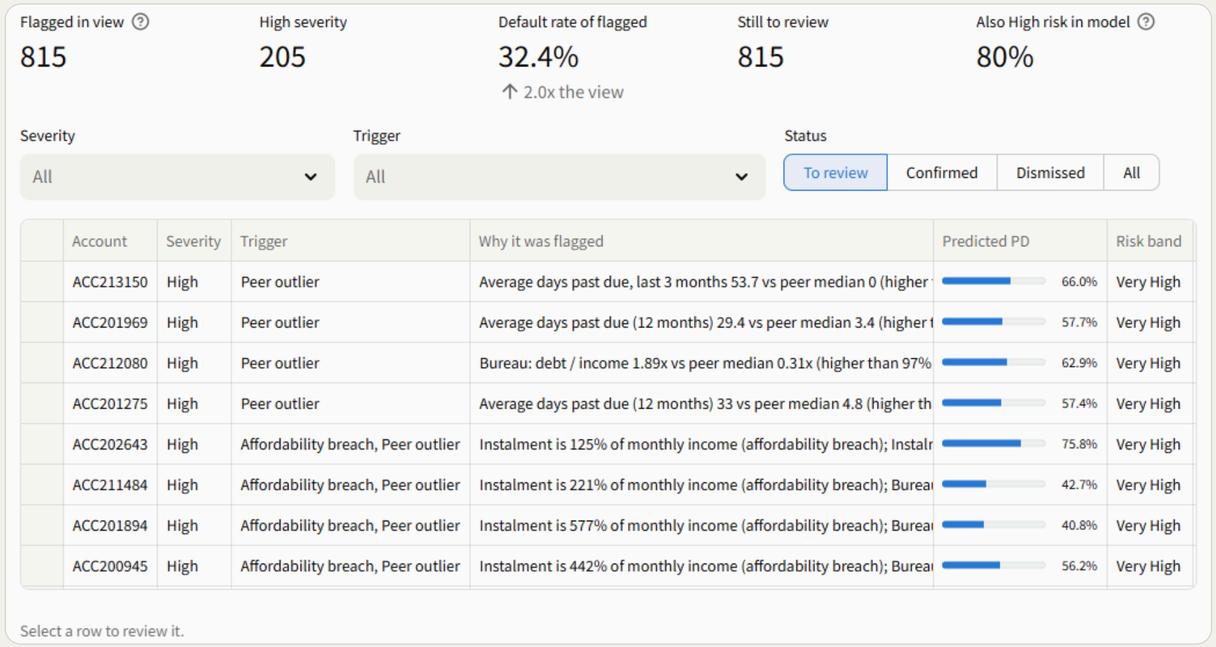
DECISIONS TODAY

Approve the pilot · ask Credit Policy to size the 40% affordability cap · sponsor access to dated bureau pulls and collateral values

What IT and data teams must provide: docs/assumptions_log.md, Part B

BONUS · BEYOND THE BRIEF

A second AI feature: anomaly review



Flags accounts that look unusual for their peer group or against their own history, with the reason in plain words.

815

accounts flagged; they default at 32.4%, twice the book

47.3%

default rate for the 205 High-severity flags

An analyst confirms or dismisses each flag; every decision is logged.

Triggers: peer outlier (Isolation Forest), instalment above income, sudden missed payments. Details: docs/ai_features.md

APPENDIX · AFTER GO-LIVE

How we keep it safe, and what it can’t do yet

Monitoring plan

Monthly: score PSI and feature CSI against the baseline (today below 0.01). Investigate above 0.10, act above 0.25.

Quarterly: AUC, KS, calibration by band and fairness ratios as outcomes mature.

Yearly: retrain, with LightGBM re-tested as the challenger.

Anomaly queue: confirm and dismiss rates tune the thresholds.

Honest limits

Synthetic data: the findings show the method, not this bank’s book.

A ceiling of about 0.70 AUC in this data; higher would mean leakage.

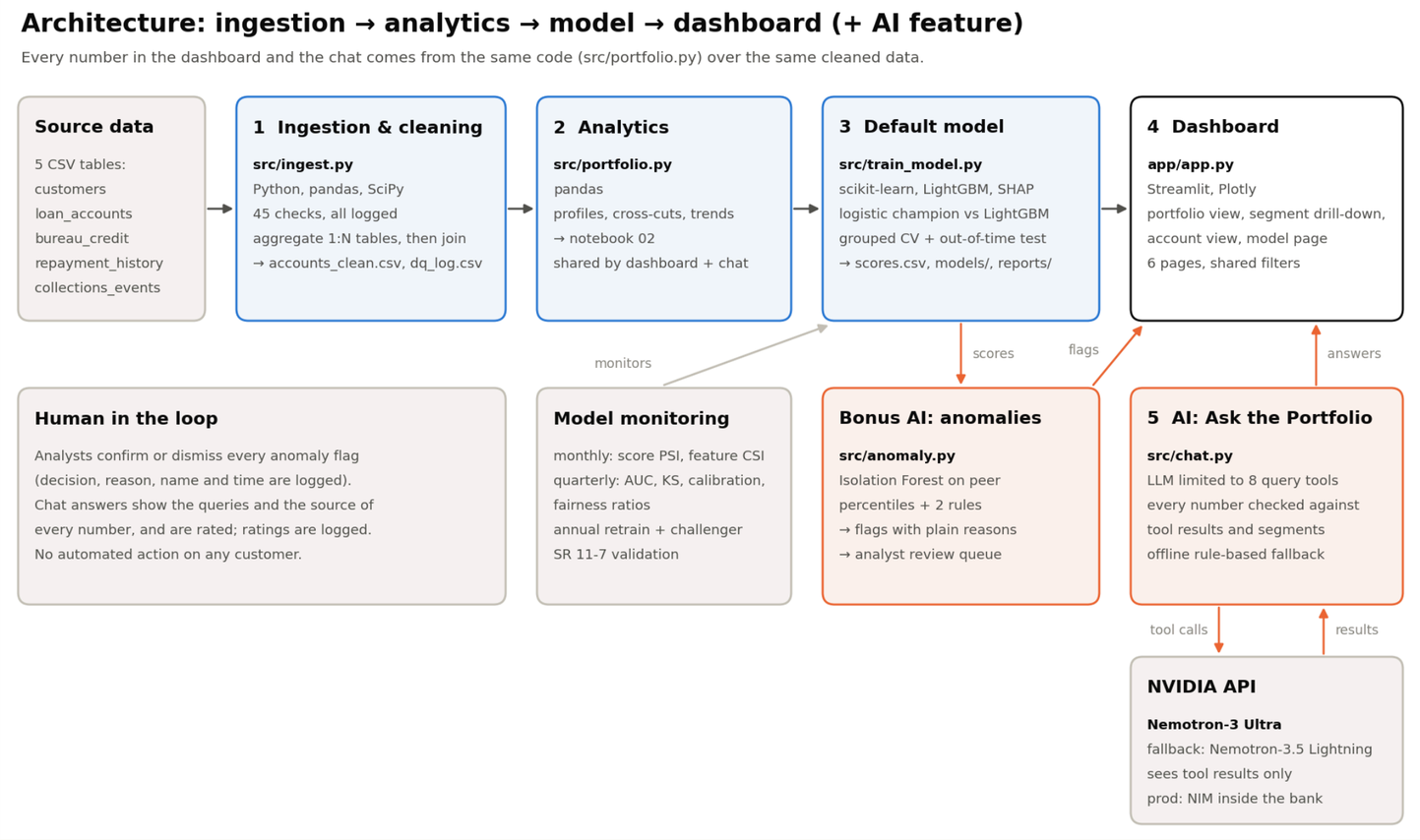
Bureau data is a snapshot; production needs pulls dated to each application.

Expected loss is illustrative: no recovery or collateral data yet.

PSI / CSI = population / characteristic stability index. Source: docs/model_report.md, docs/assumptions_log.md

APPENDIX

How it’s built: one source of numbers



Python · pandas · scikit-learn · LightGBM · SHAP · Streamlit · Plotly · NVIDIA Nemotron. Source: docs/architecture.md